URBAN NEST — DATABASE ENTITY-RELATIONSHIP DIAGRAM

MODULE 1: SAAS TENANTS, SUBSCRIPTIONS, USERS & AUTH

Multi-Tenant PG & Property Management SaaS | PostgreSQL / Supabase Schema

Tenant		Platform Root
[PK]	**id***	varchar
	name*	varchar
[UQ]	**slug***	varchar
[UQ]	**email***	varchar
	phone	varchar
	address	varchar
	city	varchar
	state	varchar
	country*	varchar
	status*	TenantStatus
	plan*	SubscriptionPlan
	subscriptionStatus*	SubscriptionStatus
	trialEndsAt	timestamp
	subscriptionStartedAt	timestamp
	subscriptionEndsAt	timestamp
	maxProperties*	integer
	maxRooms*	integer
	maxResidents*	integer
	createdAt*	timestamp
	updatedAt*	timestamp
INDEXES & CONSTRAINTS		
INDEX: (slug)		
INDEX: (status)		
INDEX: (plan)		

Plan		Pricing Model
[PK]	**id***	varchar
[UQ]	**name***	varchar
[UQ]	**tier***	SubscriptionPlan
	priceMonthly*	decimal
	priceYearly*	decimal
	maxProperties*	integer
	maxRooms*	integer
	maxResidents*	integer
	features*	varchar
	isActive*	boolean
	createdAt*	timestamp
	updatedAt*	timestamp

1 : N (tenantId)

User		Identities
[PK]	**id***	varchar
[UQ]	**email***	varchar
	passwordHash*	varchar
	name*	varchar
	role*	UserRole
	mobile	varchar
	avatarUrl	varchar
[FK]	tenantId	varchar
[FK]	propertyId	varchar
	createdAt*	timestamp
	updatedAt*	timestamp
INDEXES & CONSTRAINTS		
INDEX: (email)		
INDEX: (role)		
INDEX: (tenantId)		

ASSOCIATED ENUMS & CODES

- TenantStatus: TRIAL, ACTIVE, SUSPENDED, CANCELLED, EXPIRED, ARCHIVED
- SubscriptionPlan: TRIAL, STARTER, PROFESSIONAL, ENTERPRISE
- SubscriptionStatus: ACTIVE, TRIALING, PAST_DUE, CANCELLED, EXPIRED
- UserRole: SUPER_ADMIN, OWNER, MANAGER, RECEPTIONIST,
- ACCOUNTANT, MAINTENANCE, RESIDENT

CROSS-PAGE RELATIONSHIP CONTINUATIONS

- Property.tenantId > Tenant.id (Continues to Page 2: Property)
- User.propertyId > Property.id (Continues to Page 2: Property)
- Resident.userId - User.id (Continues to Page 2: Resident)
- AuditLog.tenantId / actorId (Continues to Page 5: AuditLog)
- SupportTicket.tenantId (Continues to Page 5: SupportTicket)

SECURITY & IDENTITY ENFORCEMENT RULES

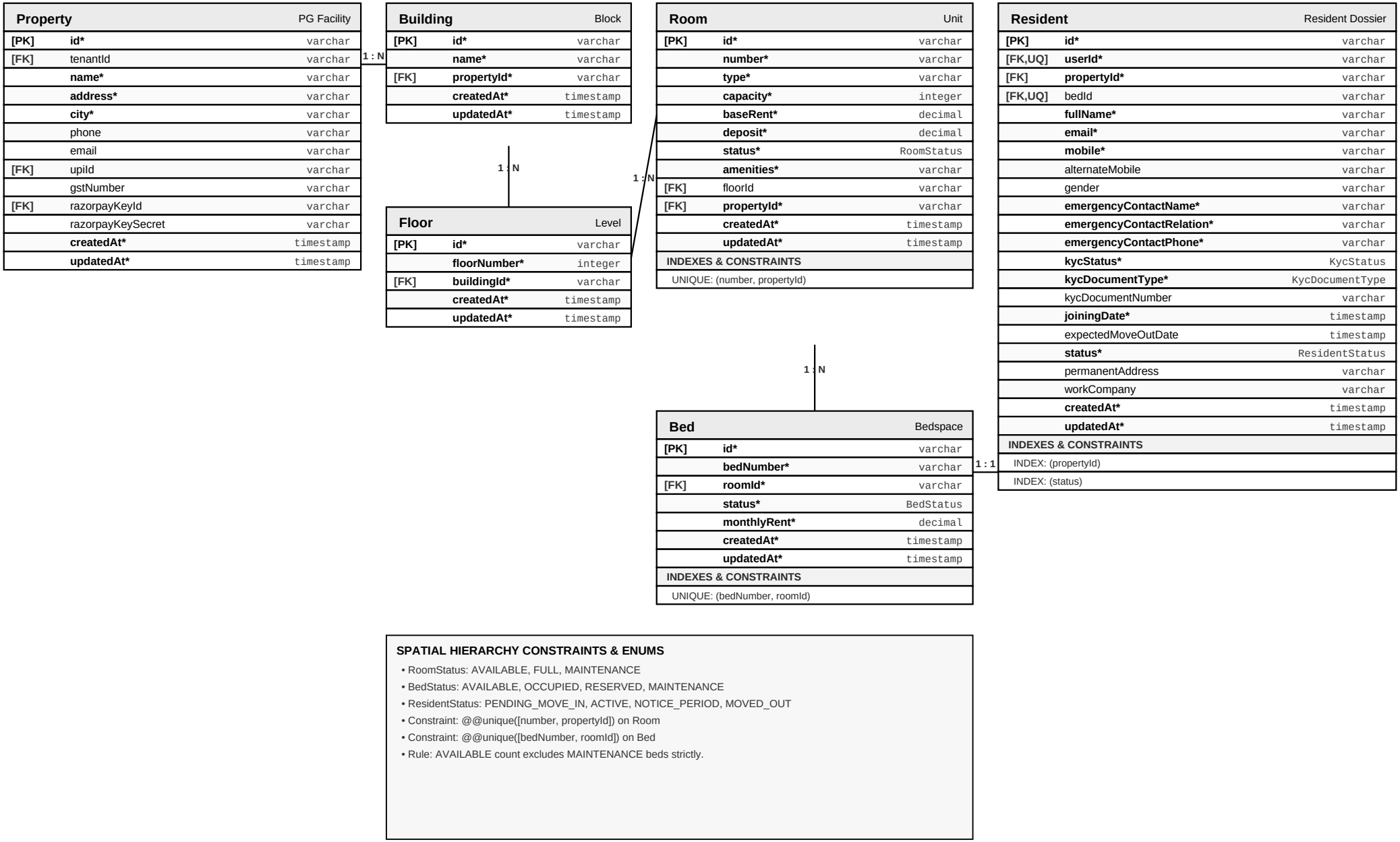
- 1. SUPER_ADMIN: Unrestricted tenant & global plan authority.
- 2. OWNER / STAFF: Strict tenant boundary via req.user.tenantId.
- 3. RESIDENT: Linked 1-to-1 with Resident dossier via userId.
- 4. PASSWORDS: Stored as bcryptjs cryptographic hash (passwordHash).
- 5. SESSIONS: Stateless JWT bearer tokens with 7-day expiration.
- 6. AUDIT: Every login, update, and deletion writes to AuditLog.

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Multi-Tenant PG & Property Management SaaS | PostgreSQL / Supabase Schema

MODULE 2: PROPERTY HIERARCHY, SPATIAL UNITS & RESIDENTS

Sheet 2 of 5 (A4 Landscape 297 x 210 mm)



SPATIAL HIERARCHY CONSTRAINTS & ENUMS

- RoomStatus: AVAILABLE, FULL, MAINTENANCE
- BedStatus: AVAILABLE, OCCUPIED, RESERVED, MAINTENANCE
- ResidentStatus: PENDING_MOVE_IN, ACTIVE, NOTICE_PERIOD, MOVED_OUT
- Constraint: @@unique([number, propertyId]) on Room
- Constraint: @@unique([bedNumber, roomId]) on Bed
- Rule: AVAILABLE count excludes MAINTENANCE beds strictly.

URBAN NEST — DATABASE ENTITY-RELATIONSHIP DIAGRAM

MODULE 3: AGREEMENTS, SECURITY DEPOSITS & FINANCIAL TRANSACTIONS

Multi-Tenant PG & Property Management SaaS | PostgreSQL / Supabase Schema

Agreement		Lease Contract
[PK]	**id***	varchar
[FK]	**residentId***	varchar
[FK]	**propertyId***	varchar
[UQ]	**agreementNumber***	varchar
	validFrom*	timestamp
	validTill*	timestamp
	signedUrl	varchar
	status*	AgreementStatus
	createdAt*	timestamp
	updatedAt*	timestamp

SecurityDeposit		Deposit Holding
[PK]	**id***	varchar
[FK]	**residentId***	varchar
[FK]	**propertyId***	varchar
	amount*	decimal
	status*	DepositStatus
	paidAt	timestamp
	refundAmount	decimal
	deductions	decimal
	notes	varchar
	createdAt*	timestamp
	updatedAt*	timestamp

Payment		Invoices & Orders
[PK]	**id***	varchar
[FK]	**residentId***	varchar
[FK]	**propertyId***	varchar
	amount*	decimal
	category*	ChargeCategory
	period*	varchar
	dueDate*	timestamp
	paidDate	timestamp
	status*	PaymentStatus
	method	PaymentMethod
[FK]	transactionId	varchar
[FK]	orderId	varchar
	notes	varchar
	createdAt*	timestamp
	updatedAt*	timestamp
INDEXES & CONSTRAINTS		
INDEX: (residentId)		
INDEX: (propertyId)		
INDEX: (status)		

1 : 1 (paymentId)

PaymentReceipt		Proof of Payment
[PK]	**id***	varchar
[FK,UQ]	**paymentId***	varchar
[UQ]	**receiptNumber***	varchar
	generatedAt*	timestamp
	pdfUrl	varchar
	createdAt*	timestamp

FINANCIAL WORKFLOW & ENUMS

- PaymentStatus: PENDING, PAID, OVERDUE, APPROVAL_PENDING, FAILED
- PaymentMethod: CASH, UPI, NET_BANKING, CARD, RAZORPAY
- ChargeCategory: RENT, SECURITY_DEPOSIT, ELECTRICITY, FOOD, MAINTENANCE_FEE, LATE_FINE, OTHER
- DepositStatus: PENDING, PAID, REFUNDED, PARTIALLY_REFUNDED
- AgreementStatus: DRAFT, ACTIVE, EXPIRED, TERMINATED
- Receipt Number: UN-REC-XXXXXX generated atomically in $transaction.
- Razorpay Flow: Order Creation -> HMAC-SHA256 Sig Verify -> Receipt

FINANCIAL CROSS-PAGE INTEGRATIONS

- Inbound References from Other Modules:
- • Resident.id -> Agreement.residentId (Leasee)
- • Resident.id -> SecurityDeposit.residentId (Depositor)
- • Resident.id -> Payment.residentId (Payee)
- • Property.id -> Agreement, SecurityDeposit, Payment (Facility scope)
-
- Idempotent Settlement Rules:
- 1. Offline collections recorded with CASH / UPI + staff audit.
- 2. Online settlement updates Payment.status = PAID and generates
- linked PaymentReceipt record atomically.
- 3. Webhook handler processes payment.captured without duplication.

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MODULE 4: GATE SECURITY, MAINTENANCE TICKETING & DOCUMENTS

Multi-Tenant PG & Property Management SaaS | PostgreSQL / Supabase Schema

VisitorRequest		Gate Pass
[PK]	**id***	varchar
[FK]	residentId	varchar
[FK]	**propertyId***	varchar
	visitorName*	varchar
	visitorMobile*	varchar
	relation*	varchar
	purpose*	varchar
	visitDate*	timestamp
	expectedEntryTime*	varchar
	expectedExitTime	varchar
	status*	VisitorStatus
	approvedBy	varchar
[UQ]	**qrPassToken***	varchar
	checkInTime	timestamp
	checkOutTime	timestamp
	createdAt*	timestamp
	updatedAt*	timestamp
INDEXES & CONSTRAINTS		
INDEX: (residentId)		
INDEX: (propertyId)		
INDEX: (status)		

Complaint		Work Orders
[PK]	**id***	varchar
[UQ]	**ticketNumber***	varchar
[FK]	**residentId***	varchar
[FK]	**propertyId***	varchar
[FK]	roomId	varchar
	title*	varchar
	description*	varchar
	category*	ComplaintCategory
	priority*	ComplaintPriority
	status*	ComplaintStatus
	assignedStaff	varchar
	resolvedAt	timestamp
	rating	integer
	feedback	varchar
	createdAt*	timestamp
	updatedAt*	timestamp
INDEXES & CONSTRAINTS		
INDEX: (residentId)		
INDEX: (propertyId)		
INDEX: (status)		

1 : N (complaintId)

MaintenanceActivity		Audit Trail
[PK]	**id***	varchar
[FK]	**complaintId***	varchar
	status*	ComplaintStatus
	updatedBy*	varchar
	comment*	varchar
	timestamp*	timestamp

Document		KYC & Files
[PK]	**id***	varchar
[FK]	**residentId***	varchar
[FK]	**propertyId***	varchar
	title*	varchar
	type*	DocumentType
	fileUrl*	varchar
	fileSize	integer
	mimeType	varchar
	status*	DocumentStatus
	rejectionReason	varchar
	uploadedAt*	timestamp
	updatedAt*	timestamp
INDEXES & CONSTRAINTS		
INDEX: (residentId)		
INDEX: (propertyId)		

LeaveRequest		Leave Applications
[PK]	**id***	varchar
[FK]	**residentId***	varchar
[FK]	**propertyId***	varchar
	fromDate*	timestamp
	toDate*	timestamp
	reason*	varchar
	status*	LeaveStatus
	approvedBy	varchar
	appliedAt*	timestamp
	updatedAt*	timestamp
INDEXES & CONSTRAINTS		
INDEX: (residentId)		
INDEX: (propertyId)		

GATE PASS & KYC ENUMS

- VisitorStatus: PENDING, APPROVED, REJECTED, CHECKED_IN, CHECKED_OUT, EXPIRED
- qrPassToken: Cryptographic UUID v4 token
- DocumentType: AADHAAR, PAN, AGREEMENT, POLICE_VERIFICATION, RECEIPT
- DocumentStatus: PENDING, VERIFIED, REJECTED
- LeaveStatus: PENDING, APPROVED, REJECTED

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MODULE 5: GOVERNANCE, OPERATIONS, INVENTORY & PLATFORM SERVICES

Multi-Tenant PG & Property Management SaaS | PostgreSQL / Supabase Schema

Sheet 5 of 5 (A4 Landscape 297 x 210 mm)

Notice — Announcements

[PK]	**id***	varchar
[FK]	**propertyId***	varchar
	title*	varchar
	content*	varchar
	category*	NoticeCategory
	isImportant*	boolean
	publishedAt*	timestamp
	expiresAt	timestamp
	publisherName*	varchar
	createdAt*	timestamp
	updatedAt*	timestamp

INDEXES & CONSTRAINTS

- INDEX: (propertyId)

Expense — Operating Cost

[PK]	**id***	varchar
[FK]	**propertyId***	varchar
	title*	varchar
	category*	ExpenseCategory
	amount*	decimal
	vendor	varchar
	date*	timestamp
	receiptUrl	varchar
	notes	varchar
	createdAt*	timestamp
	updatedAt*	timestamp

INDEXES & CONSTRAINTS

- INDEX: (propertyId)
- INDEX: (date)

AuditLog — Immutable Trail

[PK]	**id***	varchar
[FK]	tenantId	varchar
[FK]	propertyId	varchar
[FK]	actorId	varchar
	actorName*	varchar
	actorRole*	varchar
	action*	varchar
	entity*	varchar
[FK]	entityId	varchar
	timestamp*	timestamp
	ipAddress	varchar
	details*	varchar

INDEXES & CONSTRAINTS

- INDEX: (tenantId)
- INDEX: (propertyId)
- INDEX: (timestamp)

SupportTicket — Helpdesk

[PK]	**id***	varchar
[FK]	tenantId	varchar
	creatorEmail*	varchar
	creatorName	varchar
	subject*	varchar
	description*	varchar
	priority*	TicketPriority
	status*	TicketStatus
	assignedAdmin	varchar
	internalNotes	varchar
	createdAt*	timestamp
	updatedAt*	timestamp

INDEXES & CONSTRAINTS

- INDEX: (tenantId)
- INDEX: (status)

Notification — Alerts

[PK]	**id***	varchar
[FK]	**userId***	varchar
[FK]	residentId	varchar
	title*	varchar
	message*	varchar
	type*	varchar
	isRead*	boolean
	linkUrl	varchar
	createdAt*	timestamp

INDEXES & CONSTRAINTS

- INDEX: (userId)
- INDEX: (isRead)

InventoryItem — Assets & Stock

[PK]	**id***	varchar
[FK]	**propertyId***	varchar
	name*	varchar
	category*	varchar
	quantity*	integer
	minQuantity*	integer
	status*	AssetCondition
	location	varchar
	purchaseDate	timestamp
	vendor	varchar
	cost	decimal
	warrantyExpiry	timestamp
	createdAt*	timestamp

Setting — System Config

[PK]	**key***	varchar
	value*	varchar

SOSEvent — Emergency

[PK]	**id***	varchar
[FK]	**residentId***	varchar
[FK]	**propertyId***	varchar
	status*	SOSStatus
	triggeredAt*	timestamp
	acknowledgedBy	varchar
	resolvedAt	timestamp
	notes	varchar

INDEXES & CONSTRAINTS

- INDEX: (residentId)
- INDEX: (propertyId)

OperationalTask — Staff Duty

[PK]	**id***	varchar
[FK]	**propertyId***	varchar
	title*	varchar
	category*	varchar
	priority*	TaskPriority
	assignedTo	varchar
	dueDate	timestamp
	status*	TaskStatus
	notes	varchar
	createdAt*	timestamp
	updatedAt*	timestamp

INDEXES & CONSTRAINTS

- INDEX: (propertyId)

OPERATIONAL ENUMS

- NoticeCategory: GENERAL, MAINTENANCE, PAYMENT, EMERGENCY, EVENT, RULES
- SOSStatus: ACTIVE, ACKNOWLEDGED, RESOLVED
- AssetCondition: EXCELLENT, GOOD, NEEDS_REPAIR, REPLACED, LOW_STOCK
- TaskStatus: PENDING, IN_PROGRESS, COMPLETED
- TicketStatus: OPEN, RESOLVED

Legend: [PK] Primary Key | [FK] Foreign Key | [UQ] Unique | ---> Many-to-One | --- One-to-One | * Required field

Academic & Production Standard | Source: server/prisma/schema.prisma